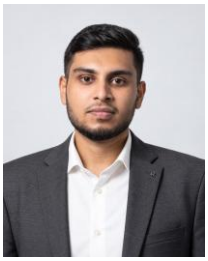
RAYAN ALAM



Robotics Software Engineer | Computer Vision & Autonomous Systems
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PROFESSIONAL SUMMARY

Robotics Software Engineer and Co-Founder & Lead Researcher at SynthaNano Research Lab, focused on autonomous navigation, machine learning, and computer vision. Projects span ROS 2/Nav2 robots, YOLO detection, medical-image classification, and NLP clustering. Experienced in robotics team leadership.

PROFESSIONAL EXPERIENCE

Co-Founder & Lead Researcher | September 1, 2026 - Present

SynthaNano Research Lab | www.synthananolab.com

Robotics Software Engineer | May 10, 2026 - Present

Nyza Engineering Design

Backend Developer | March 1, 2025 - October 31, 2025

Valencia Air Travels & Tours Ltd

TECHNICAL SKILLS

Programming: Python, PHP, C++, C#, Java

Robotics and Embedded Systems: ROS, ROS 2, Nav2, SLAM, A* planning, lidar, odometry, IMU fusion, Jetson, Raspberry Pi, Arduino, NodeMCU

Machine Learning and Computer Vision: PyTorch, EfficientNet-B0, YOLO, CNNs, model training, custom datasets, cross-validation, image processing

Backend and Web: Laravel, Flask, API integration, MySQL, HTML, CSS

NLP and Unsupervised Learning: TF-IDF, LSA, UMAP, K-means, HDBSCAN, Doc2Vec

Mobile and Core Concepts: Flutter, data structures, algorithms

SELECTED PROJECTS

Breast Ultrasound Lesion Classification | PyTorch, EfficientNet-B0

- Co-authored an EfficientNet-B0 study comparing full-image and mask-derived inputs for benign/malignant classification across three ultrasound datasets.
- Used grouped cross-validation, ablation studies, calibration, and bootstrap uncertainty analysis.

LLM and Human-AI Research Clustering | TF-IDF, LSA, UMAP

- Co-authored a K-means/HDBSCAN analysis of 10,000 arXiv abstracts covering LLM and human-AI interaction research.
- Evaluated stability and coherence; mapped 14 themes with five shared clusters at a 70% dominance threshold.

Dog Waste Cleaning Robot | ROS 2, Nav2, Jetson, YOLO26n

- Trained YOLO26n on a custom dataset of 1,500 images; integrated detection on NVIDIA Jetson with OAK-D Pro colour and depth streams.
- Integrated ROS 2/Nav2 waypoint navigation and A* planning with RPLidar S2 localization, wheel odometry, and BNO055 IMU heading fusion.
- Implemented depth-guided target approach and brush-control sequences, with a web dashboard for camera/depth views, manual driving, saved waypoints, and RPM/IMU telemetry.

Autonomous Factory Robot | ROS, Raspberry Pi, SLAM

- Developed a ROS/Raspberry Pi robot with SLAM navigation, obstacle avoidance, and mobile-app autonomous/manual control for material transport.

Aerial Crop Yield Prediction | Drones, YOLO

- Captured aerial rice-field footage, created a custom dataset, and trained YOLO to detect rice panicles and support yield prediction.

Drone-Based Weapon Detection | YOLO, Cloud Inference

- Built cloud-deployed YOLO inference with live drone video streaming for weapon detection.

Smart Blind Assistant | Python, YOLO, Flask, Flutter, GPS

- Integrated YOLO object detection, a Flask API, and Flutter speech output in smart glasses; added smart-stick obstacle alerts and GPS location tracking.

IoT Smart Home System | Arduino, NodeMCU, Flutter

- Built Arduino/NodeMCU door access, gas/fire alerts, and automatic watering with Flutter remote control.

Crowdfunding Web Application | PHP, MySQL, HTML, CSS

- Developed a crowdfunding website using PHP, MySQL, HTML, and CSS.

EDUCATION

Bachelor of Science in Computer Science and Engineering
American International University-Bangladesh (AIUB), Dhaka
September 2021 - Present | CGPA: 3.66/4.00

Higher Secondary Certificate (HSC), Science
Rajbari Govt. College, Rajbari Sadar, Rajbari
2019 - 2020 | GPA: 4.33/5.00

Secondary School Certificate (SSC), Science
Rajbari Govt. High School, Rajbari Sadar, Rajbari
2017 - 2018 | GPA: 4.39/5.00

SELECTED ACHIEVEMENTS

- Gold Medal, Technology - Global Youth Invention and Innovation Fair 2026, Indonesia.
- Gold Medal and IYSA Semi Grand Award - WYIIA 2025, Surabaya, Indonesia; Team Leader.
- Champion - 7th WICE 2025 Bangladesh National Round; selected for Malaysia international round.
- Second Runner-Up, Project Showcasing - Robolution 2025, MIST Robotics Club.
- Second Runner-Up, Robo Soccer - AIUB CS Fest 2024.

LEADERSHIP AND ACTIVITIES

Team Co-Lead and AI and Autonomous Sub-Team Lead - Red Frontier Mars Rover Team
Software Team Crew Member - AIUB Robotic Crew

CERTIFICATION

IT Essentials: PC Hardware and Software - Cisco Networking Academy, 2021

REFERENCE

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American International University-Bangladesh (AIUB)
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